Sivakumaran Sarvanan

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Google Scholar: scholar.google.com/citations?user=Jnko-ZcAAAAJ

Education

- **DSTI School of Engineering, France** 2024 – 19 March 2026
  MSc in Data Science & AI (First Class / Summa Cum Laude); Thesis: *Enhancing Medical QA with Qwen3-0.6B on MedQuAD*
- **University of Jaffna, Sri Lanka** 2018 – 2022
  BSc (Hons) in Computer Science (Second Class Upper); Thesis: *Leaf Disease Detection on Brinjal Using ML*

Awards & Scholarships

- DSTI – Fondation DST Scholarship, 2025 (partial), including M1 Merit Scholar
- Mahapola Scholarship, University of Jaffna
- Australian Engineering Circle Scholar

Part-Time Tutor — VKeerthi Academy

Basic Computer Skills and Data Science Tutor Part-Time

- Teach basic computer skills, including computer fundamentals, file management, and common software applications.
- Introduce students to data science concepts, data analysis, and Python programming.
- Prepare learning materials and practical exercises to support students' understanding of computer science and data science fundamentals.
- Explain technical concepts clearly and guide students through hands-on exercises and problem-solving activities.

Student Talent — DataLens.Tools

Python, Data Science and AI Learning Support Oct 2026 – Present
Ongoing student talent role involving technical learning, Python programming, data science and AI-related problem-solving, including debugging, understanding technical concepts, reviewing solutions, and responsible use of AI-assisted programming tools.

Business / Industry Project — Qnology AI Ltd

Text Mining and Stock Market Pattern Recognition Ongoing
Proposed remote research/project collaboration focusing on financial text mining, market-data analysis, sentiment and event detection, and machine-learning-based stock-market pattern recognition.

Research Experience

- **MSc Intern – HarmoniaHealthAI, Johannes Gutenberg Univ., Germany** Aug 2025 – Feb 2026
  Developed NLP and LLM models for medical chatbots, fine-tuning, dataset engineering, and performance evaluation.
- **MSc Thesis – Enhancing Medical QA with QWEN-3 0.6B, DSTI, Paris** 2025 – 2026
  Fine-tuned Qwen3-0.6B on MedQuAD using LoRA and 4-bit quantization; evaluated using BLEU, ROUGE-L, F1, METEOR.
- **BSc Project – Brinjal Leaf Disease Detection, Univ. of Jaffna** 2021 – 2022
  Image preprocessing, PCA/HOG feature extraction, KNN classification, clustering, and probabilistic optimization.

Publications

Poster & Abstracts

- "Identify the Leaf Diseases on Brinjal Using Machine Learning" – FARS2022 Proceedings

Journal Articles

- **"A Specific Model Using Leaf Detection Outperforms Deep Learning Models"**, IEEE-SEM, Vol. 11, Issue 8 (August 2023).
  Click here to view
- **"Automated Brinjal Leaf Disease Detection Using Machine Learning: A Sri Lankan Study"**.
  Click here to view PDF
- **"Deep Learning–Based Land Use and Land Cover Classification Using the EuroSAT Dataset"**.
  Click here to view

Scholarly Submissions / Not Yet Published

- **Sarvanan, S.** (2026). *Unsupervised Anomaly Detection and Clustering of Kidney Stone CT Images Using Hybrid Feature Extraction.* Submitted to the **Trustworthy Document Understanding: Privacy, Unlearning, Robustness, and Explainability Workshop**, ICPR 2026, International Convention Center, Lyon, France. Date: August 21, 2026. Status: Under review.
- **Sarvanan, S.** (2026). *Enhancing Medical Question Answering with Qwen3-0.6B Using the MedQuAD Dataset.* Submitted to ALLIES MSCA COFUND. Status: Under review.
- **Sarvanan, S.** (2026). *Hybrid Unsupervised Feature Learning for Anomaly Detection and Clustering in Kidney CT Imaging.* Accepted abstract and poster, International Conference on Anomaly Detection in Industrial Systems Using Data Science (ICADISS-26), Zurich, Switzerland, 29–30 September 2026. Article ID: ISER-2-25-0049492.
  Academia Profile

Manuscripts in Preparation / Unpublished Work

- Sivakumaran Sarvanan, *et al.* **Computer Vision Approaches for Economic Activity Estimation from Satellite Imagery: A State-of-the-Art Review**.
  Technical report prepared with colleagues; manuscript in preparation.
- Sivakumaran Sarvanan, *et al.* **Unified Multi-Modal Explainable AI Interface for Audio and Image Classification**.
  Software-focused research project integrating explainable AI techniques; manuscript in preparation.

Research Interests

Machine Learning; Unsupervised Self-Supervised Learning; Pattern Recognition; Representation Learning; Anomaly Detection; Clustering; Probabilistic Modeling; Generative AI; Multimodal Learning; Healthcare AI; Explainable AI; Signal Processing.

Selected Projects

- ODD Protocol-Based Agent: Market Basket Analysis
- Book Recommendation System (Goodreads)
- Mountain Car RL (Reinforcement Learning)
- Prohibited Comments Classifier (NLP)

Technical Skills

Python, R, SQL, Java, C++, PHP; Scikit-learn, TensorFlow, PyTorch, Keras, PCA, Clustering, NLP; Git, Flask, MySQL, Neo4j, Google Colab, NetLogo

Teaching & Work Experience

- Teaching Assistant, Univ. of Jaffna (2022 – 2024): Assisted ML/Data Science labs, mentored students
- BSc Intern – HelloFoodies, India (Feb 2022 – Jul 2022): Flask web app, delivery optimization, clustering, predictive analytics

Volunteer Work

- Volunteer Maths Teacher, Australian Engineering Circle (2016 – 2024)
- Prospective NextGen@ICANN86 Participant (Present)

Workshops / Conference Participation

- Professional Development — Climate Data Understanding Climate Data Ongoing
  European Union Programme with Wageningen University & Research
  Currently completing a self-paced course focused on understanding, interpreting, and working with climate data, including key concepts in climate datasets, analysis, and data-driven approaches to climate-related research.

- Edge AI Vision Summit 2026 – Workshop participation on Edge AI and computer vision systems, organized by the Edge AI and Vision Alliance, Santa Clara, CA, USA (registration)
- Member of the French Association for Pattern Recognition (AFRIF).
- FARS2022 – International workshop/conference participation, Sri Lanka.
- SLSWCS – Sri Lanka Symposium/Workshop on Computer Science participation, Sri Lanka.

Competitions / Shared Tasks

- MEDIQA-EVAL @ ClinicalNLP 2026, multimodal medical QA evaluation; part of MEDIQA Shared Tasks Participation ongoing (website)
- Image Detection Challenge @ AI4Media Benchmark 2026, participation in image manipulation detection task (website)

Certifications

Neo4j Certified Professional (2024); Write & Publish a Scientific Article – Inria (2024)

Languages

Tamil (Native), English (Fluent), French (A2, Université Paris Cité, Faculty of Humanities and Social Sciences)

References

Available upon request.